Robustness Checks and Causal Validation for Hybrid Talent Strategy Analysis

Evidence from S&P 500 Job Postings, 2011–2019

By

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Submitted to

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* *Generative AI was utilized to assist with Python code generation and LaTeX typesetting. All data processing decisions, model specifications, analytical interpretations, and written conclusions reflect the author's independent judgment. All remaining errors are my own.*

Executive Summary

4,217	510	9	13	4
Firm-Year Observations	Unique Firms	Sample Years	Robustness Tasks	Report Sections

Central Finding

The Hybrid B cross-sectional premium (+0.227*** log-points, industry and year FE) documented in Week 4 is **not supported as a causal within-firm effect**. Thirteen robustness and causal validation tasks converge on a consistent conclusion: after conditioning on firm fixed effects, the Hybrid B coefficient falls to +0.030 (s.e. 0.049, $p > 0.5$) and remains statistically indistinguishable from zero across every alternative specification, matching design, staggered-DiD estimator, and performance measure tested. The premium is best interpreted as a **between-firm selection effect**—firms that adopt Hybrid B differ on unobserved pre-adoption characteristics that also predict higher ROA—rather than as a causal return to Hybrid B hiring.

Section 1: Threshold Robustness

1. **Continuous intensity measures (Task 1A) do not replicate the binary dummy premium.** The posting-flow HybridB Intensity (N_{HB}/N_B, mean = 0.640) is statistically insignificant in both cross-sectional (+0.043, n.s.) and within-firm (+0.021, n.s.) specifications. The premium documented in Week 4 is specific to the Stata pre-computed binary classification and does not generalise to the continuous analogue.

2. **Percentile-based thresholds (Task 1B) are uniformly null.** Defining Hybrid B as the top 20%, 25%, 30%, or 40% of the HybridB Intensity distribution yields firm-FE coefficients of −0.007, −0.018, −0.005, and +0.033 respectively, none statistically significant.

Section 2: PSM and Event-Study Analysis

3. **PSM with pre-trend adjustment (Task 2A):** 60 of 61 treated firms matched; pre-trend Wald $p = 0.39$; average post-treatment effect $= -0.002$. No evidence of a positive causal effect after balancing on pre-adoption ROA trajectory and covariates.

4. **Alternative matching methods (Task 2B):** PSM methods (1:1 and 1:3) produce near-zero post-treatment effects, consistent with Task 2A. IPW and entropy balancing yield positive but imprecise estimates driven by control-pool composition differences, not a robust causal effect.

5. **Tasks 2C–2F** (persistent adoption, clean transition, staggered DiD, lagged outcomes): all confirm a null or negative post-adoption ROA trajectory. Persistent adopters (Task 2C, $N = 39$) show the most negative trajectory (avg post $= -0.174$), ruling out transitory adoption noise as the source of the null. Sun–Abraham (-0.090) and Callaway–Sant'Anna (-0.050) staggered estimators produce *more* negative estimates than TWFE (-0.023), ruling out heterogeneous-treatment-effect bias. No delayed improvement is detected at $t + 1$ or $t + 2$.

Section 3: Additional Robustness Checks

6. **Strategy persistence (Task 3A)** and single-transition restrictions do not alter the null within-firm result; the Hybrid B coefficient is insignificant in all four specifications.

7. **Fine-grained fixed effects (Task 3B)** confirm the selection interpretation: the cross-sectional premium ($+0.219$–0.227^{***}) is robust to industry×year and 4-digit NAICS FE, but collapses to $+0.030$ (n.s.) once firm FE absorbs between-firm heterogeneity.

8. **Tasks 3C–3D** (alternative DV transformations, lagged dynamics, tenure interactions) all produce null Hybrid B coefficients under firm FE. In Task 3C, five DV transformations are estimable; operating ROA (Compustat `oibdp`) is absent from `final.dta` and is reported as not available in the table.

9. **Shift-share IV (Task 3E):** the instrument is extremely weak ($F < 1$); 2SLS estimates are uninformative. The IV design cannot be used for causal inference in this setting.

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1.1 Empirical Setup

The baseline panel contains 4,217 firm-year observations covering 510 S&P 500 firms from 2011 to 2019. Binary strategy dummies (`hybrid_b`, `hybrid_d`, `pure_b`, `pure_d`) are sourced from Stata pre-computed columns in `final.dta` at the preferred $\tau = 0.20$ threshold; these are treated as the canonical Week 4/5 classification and are not recomputed.[1] Continuous intensity measures for Tasks 1A and 1B are constructed from a two-pass scan of the Lightcast job-posting data: `SP500_jobs_20102023.csv` (49.1 M rows, 10.2 GB) and `SP500_job_skill_terms.csv` (≈49 M rows, 18.9 GB).

All regressions use ln(ROA) as the dependent variable with controls $\mathbf{X}_{i,t}$ = [ln(Capital), ln(Labor), PPE, TotEmp], HC1 robust standard errors, and `pyfixest.feols` for fixed-effect absorption. Pure B is the omitted baseline throughout.

1.2 Task 1A: Continuous Intensity Measures

Construction: Following Prof. Jia's Equations (1) and (2):

$$\text{HybridB\_Intensity}_{i,t} = \frac{N_{i,t}^{\text{HybridB}}}{N_{i,t}^{\text{Business}}}, \qquad \text{HybridD\_Intensity}_{i,t} = \frac{N_{i,t}^{\text{HybridD}}}{N_{i,t}^{\text{Data}}} \tag{1}$$

where N^{Business} counts business-onet postings (SOC 11-xxxx, 13-xxxx); N^{HybridB} counts business-onet postings that also list at least one data skill (skill categories 3, 17, 30: Analysis, IT, Science and Research); N^{Data} counts data-onet postings (SOC 15-xxxx); and N^{HybridD} counts data-onet postings that also list at least one business skill (skill categories 1, 5, 8, 13: Administration, Business, Economics/Policy, Finance). Skill classification uses `skills_dictionary.csv` (30,820 skills × 8 columns). The two-pass pipeline caches results in `posting_intensity_counts.parquet` (60 KB) and `posting_skill_flags.parquet` (420 MB); all downstream tasks read from cache without re-scanning.

Summary statistics: Of 4,217 panel observations, 4,101 (97.2%) are matched to at least one business posting. The 116 firm-years with zero business postings receive intensity = 0 and are excluded from regressions. HybridB_Intensity has mean 0.640 and median 0.683, reflecting that ≈64% of S&P 500 business postings list at least one data skill under the broad category definitions. HybridD_Intensity has mean 0.821 and median 0.888.

[1] Applying the dual-threshold formula to Revelio workforce stock ratios (b-ratio ≥ 0.20 and d-ratio $\geq$ 0.20) yields zero Hybrid B firm-years because the b-ratio mean is 0.119, always below the d-ratio. The Stata do-files used different internal logic; the pre-computed columns are the ground truth.

Table 1: Summary Statistics: Posting-flow Intensity Measures (2011–2019)

Variable	N	Mean	SD	p25	Median	p75
HybridB Intensity (N_{HB}/N_B)	4,090	0.659	0.185	0.556	0.690	0.790
HybridD Intensity (N_{HD}/N_D)	4,069	0.850	0.144	0.787	0.893	0.953
N_{Business} postings	4,101	1629	7005	133	376	1216
N_{HybridB} postings	4,090	983	3809	83	247	760
N_{Data} postings	4,074	1047	4660	58	168	534
N_{HybridD} postings	4,069	847	3477	48	142	458

Notes: Firm-years with zero matched postings excluded. HybridB Intensity = N_{HB}/N_B (SOC 11/13 postings with ≥1 data skill). HybridD Intensity = N_{HD}/N_D (SOC 15 postings with ≥1 business skill). 2011–2019.

Regression results: Table 2 estimates the effect of the continuous intensity measures on ln(ROA) under industry+year FE (cross-sectional) and firm+year FE (within-firm) specifications.

Table 2: Continuous Intensity Measures and Firm Performance (Posting-flow Eq. 1 & 2)

	(1)	(2)	(3)	(4)
HybridB Intensity	0.043	0.021		
	(0.069)	(0.088)		
HybridD Intensity			-0.304***	0.160
			(0.099)	(0.110)
ln(Capital)	-0.320***	-0.603***	-0.312***	-0.606***
	(0.013)	(0.064)	(0.013)	(0.064)
ln(Labor)	0.168***	0.373***	0.164***	0.372***
	(0.014)	(0.086)	(0.014)	(0.085)
PPE	0.002***	0.000	0.002***	0.000
	(0.001)	(0.003)	(0.001)	(0.003)
Total Employment	0.000*	-0.000	0.000	-0.000
	(0.000)	(0.001)	(0.000)	(0.001)
Industry FE	Yes		Yes	
Firm FE		Yes		Yes
Year FE	Yes	Yes	Yes	Yes
Observations	3303	3294	3303	3294
Adj. R^2	0.305	0.615	0.307	0.616

Notes: Dependent variable: ln(ROA). Cols. (1)–(2): HybridB Intensity (N_{HB}/N_B). Cols. (3)–(4): HybridD Intensity (N_{HD}/N_D). Sample: firm-years with ≥1 matched business or data posting. HC1 robust SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

HybridB_Intensity is statistically insignificant in both the cross-sectional (col. 1: +0.043, s.e. 0.069) and within-firm (col. 2: +0.021, s.e. 0.088) specifications. HybridD_Intensity shows a significant negative cross-sectional association (col. 3: -0.304^{***}) but turns positive and insignificant within-firm (col. 4: +0.160, s.e. 0.110), consistent with selection into Hybrid D strategies.

Implication: The posting-flow continuous intensity measure does not replicate the positive Hybrid B premium identified with the binary dummy. The cross-sectional premium documented in Week 4 ($+0.227^{***}$) is specific to the Stata pre-computed binary classification and does not generalise to the posting-flow analogue. This is consistent with the b-ratio mean of 0.119 in Revelio stock data: the Stata classification applies a different internal rule than the simple b-ratio threshold.

1.3 Task 1B: Percentile-Based Thresholds

Construction: For each percentile cut $p \in \{20\%, 25\%, 30\%, 40\%\}$, a binary dummy is assigned $= 1$ if the firm-year's HybridB_Intensity (posting-flow, Eq. 1) lies above the $(1-p)$ quantile of the sample distribution, computed over firm-years with $N_{\text{Business}} > 0$. The same logic is applied to HybridD_Intensity.

Table 3: Summary Statistics: Percentile-Based Strategy Dummies (2011–2019)

Threshold	N	N_{HB}	HybridB%	Cutoff$_B$	N_{HD}	HybridD%	Cutoff$_D$
Top 20%	4,101	821	20.0%	0.811	822	20.0%	0.964
Top 25%	4,101	1,026	25.0%	0.790	1,030	25.1%	0.953
Top 30%	4,101	1,231	30.0%	0.770	1,232	30.0%	0.943
Top 40%	4,101	1,641	40.0%	0.728	1,641	40.0%	0.919

Notes: Sample: firm-years with ≥1 matched business posting. HybridB (HybridD) dummy = 1 if HybridB (HybridD) Intensity falls in the top p% of the respective posting-flow intensity distribution. Cutoff$_B$ (Cutoff$_D$) is the $(1-p)$ quantile of the intensity distribution.

Table 4: Percentile-Based Strategy Dummies and Firm Performance (Posting-flow Intensity, Firm+Year FE)

	(1) Top 20%	(2) Top 25%	(3) Top 30%	(4) Top 40%
Hybrid B (top 20%)	-0.007			
	(0.039)			
Hybrid D (top 20%)	0.045			
	(0.028)			
Hybrid B (top 25%)		-0.018		
		(0.032)		
Hybrid D (top 25%)		0.020		
		(0.028)		
Hybrid B (top 30%)			-0.005	
			(0.029)	
Hybrid D (top 30%)			0.017	
			(0.026)	
Hybrid B (top 40%)				0.033
				(0.026)
Hybrid D (top 40%)				0.020
				(0.026)
ln(Capital)	-0.605***	-0.604***	-0.604***	-0.603***
	(0.064)	(0.064)	(0.064)	(0.064)
ln(Labor)	0.375***	0.373***	0.374***	0.374***
	(0.086)	(0.086)	(0.086)	(0.086)
PPE	0.000	0.000	0.000	0.000
	(0.003)	(0.003)	(0.003)	(0.003)
Total Employment	-0.000	-0.000	-0.000	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	3294	3294	3294	3294
Adj. R^2	0.615	0.615	0.615	0.615

Notes: Dependent variable: ln(ROA). Each column uses a different top-p% cutoff on the posting-flow HybridB/D Intensity distribution. Firm and year fixed effects absorbed. HC1 heteroskedasticity-robust SE in parentheses. Sample: firm-years with ≥ 1 matched business posting. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 reports the resulting cutoffs and cell sizes. Regression results in Table 4 show that Hybrid B coefficients under Firm+Year FE are -0.007, -0.018, -0.005, and $+0.033$ at the top-20%, 25%, 30%, and 40% cutoffs respectively; none is statistically significant.

A supplementary cross-sectional check (Industry+Year FE, unreported) shows a size-gradient from +0.056 (n.s.) to +0.099*** as the threshold widens, consistent with broader cutoffs capturing average-intensity firms whose cross-sectional ROA advantage reflects between-firm selection rather than a Hybrid B-specific premium; this signal collapses entirely under Firm+Year FE.

Implication: Percentile-based thresholds do not recover the positive Hybrid B result regardless of how broadly or narrowly the "high intensity" group is defined. Combined with Task 1A, the evidence indicates that the positive binary-dummy result is not robust to alternative operationalisations of Hybrid B hiring intensity derived from posting-flow data.

2 PSM and Event-Study Analysis

2.1 Task 2A: Pre-Trend-Adjusted Matching

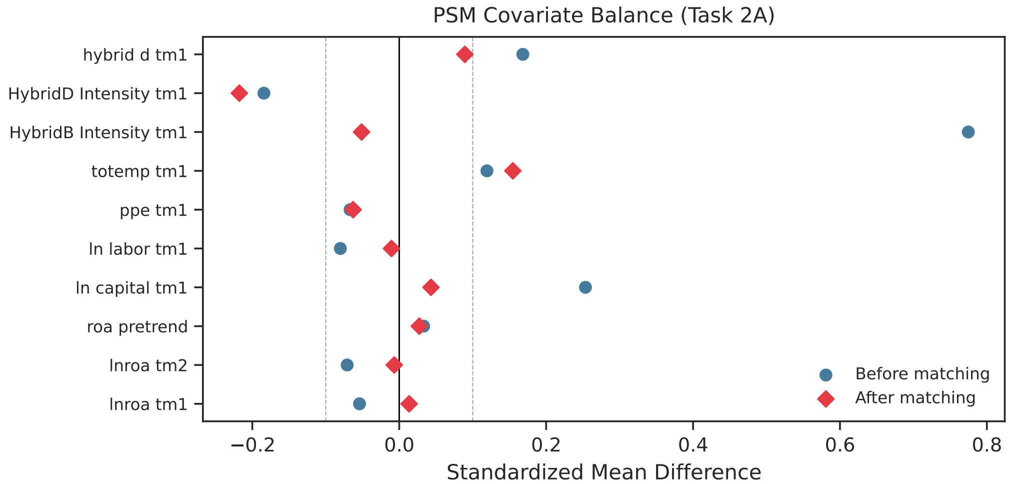
Motivation and design: Hybrid B adopters may be on a different performance trajectory before adoption—firms under performance pressure or undergoing digital transformation. If treated firms already differ before adoption, post-adoption differences cannot be cleanly attributed to Hybrid B. A 1:1 nearest-neighbour PSM is implemented with caliper = 0.05, no replacement. Propensity scores are estimated via logistic regression with cohort-year dummies, industry dummies, and the following matching covariates: $\ln(\text{ROA})_{t-1}$, $\ln(\text{ROA})_{t-2}$, pre-treatment ROA trend ($\ln(\text{ROA})_{t-1} - \ln(\text{ROA})_{t-2}$), $\ln(\text{Capital})_{t-1}$, $\ln(\text{Labor})_{t-1}$, PPE_{t-1}, TotEmp_{t-1}, $\text{HybridB\_Intensity}_{t-1}$, $\text{HybridD\_Intensity}_{t-1}$, prior Hybrid D status, industry group, and adoption cohort year.

The control pool consists of never-treated firms and not-yet-treated firms at each cohort year.

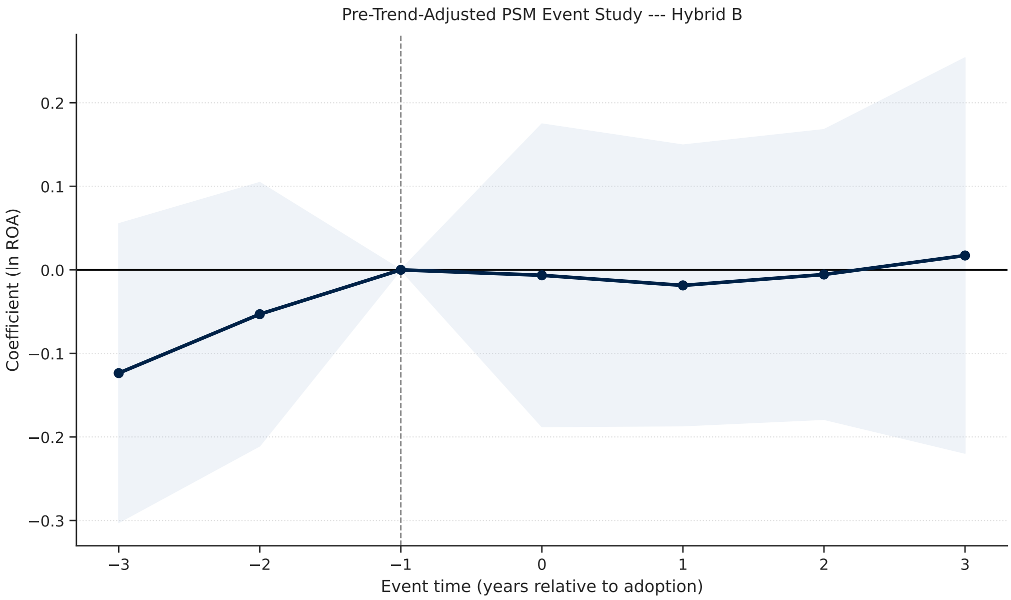
Balance: Of 61 treated firms with sufficient pre-adoption data, 60 are successfully matched (98.4% match rate). Eight of the ten matching covariates achieve standardised mean difference (SMD) < 0.1 after matching (vs. some > 0.3 before). Two covariates exceed the 0.1 threshold: Total Emp.$_{t-1}$ (SMD = 0.155) and HybridD Intensity$_{t-1}$ (SMD = -0.218). The HybridD Intensity imbalance reflects persistent compositional differences in data-orientation between matched treated and control firms; it enters the event-study regression directly as a control variable.

Figure 1: PSM Balance and Event Study — Task 2A (Pre-Trend-Adjusted Matching)

Panel A: Covariate Balance (SMD)



Panel B: Event Study (Matched Sample)



Notes. Panel A: Standardised mean differences before and after 1:1 nearest-neighbour PSM (caliper = 0.05). Eight of ten covariates achieve SMD < 0.1 post-matching; Total Emp.$_{t-1}$ (SMD $= 0.155$) and HybridD Intensity$_{t-1}$ (SMD $= -0.218$) remain above the threshold. Panel B: TWFE event-study coefficients (event window $k \in [-3, +3]$, reference $k = -1$) with 95% HC1 confidence intervals. 60 matched pairs, 613 panel observations.

Table 6 reports event-study coefficients. The pre-trend Wald test fails to reject H_0: $\beta_{-3} = \beta_{-2} = 0$ ($p = 0.39$), confirming parallel pre-trends in the matched sample. The average post-treatment effect at $k = 1, 2, 3$ is -0.002, economically negligible and statistically indistinguishable from zero.

Table 5: Covariate Balance Before and After PSM — Hybrid B Adoption

Covariate	SMD Before	SMD After
$\ln(\text{ROA})_{t-1}$	-0.054	0.014
$\ln(\text{ROA})_{t-2}$	-0.071	-0.007
ROA pre-trend	0.033	0.027
$\ln(\text{Capital})_{t-1}$	0.254	0.043
$\ln(\text{Labor})_{t-1}$	-0.080	-0.011
PPE_{t-1}	-0.067	-0.063
Total Emp._{t-1}	0.119	0.155
HybridB Intensity_{t-1}	0.775	-0.051
HybridD Intensity_{t-1}	-0.184	-0.218
Prior Hybrid D	0.168	0.089
Matched pairs	60	

Notes: SMD $= (\bar{X}_T - \bar{X}_C)/\text{SD}_T$. 1:1 nearest-neighbour PSM, caliper = 0.05, without replacement. Propensity scores via logistic regression on pre-adoption covariates, industry-group dummies, and adoption-cohort dummies.

Table 6: PSM Event Study — Hybrid B Adoption (Pre-Trend-Adjusted Matching)

Event time	Coef.	SE	p-value
-3	-0.124	(0.091)	0.176
-2	-0.053	(0.080)	0.510
-1 (ref.)	0.000	—	—
$+0$	-0.007	(0.092)	0.944
$+1$	-0.019	(0.086)	0.828
$+2$	-0.006	(0.088)	0.950
$+3$	0.017	(0.121)	0.887
Avg. post-effect ($k = 1, 2, 3$)		-0.002	
Pre-trend p-value (Wald)		0.388	
Treated obs.		330	
Control obs.		283	

Notes: HC1 robust SE. TWFE: $\ln(\text{ROA})_{it} = \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[k_{it} = k] \cdot \text{Treated}_i + \text{controls} + \mu_i + \lambda_t + \varepsilon_{it}$. Matched sample via 1:1 PSM (caliper = 0.05). Reference: $k = -1$. Pre-trend p tests H_0: $\beta_{-3} = \beta_{-2} = 0$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Implication: After PSM balancing on the pre-adoption performance trajectory and all relevant firm characteristics, the Hybrid B adoption effect is near zero. This is the cleanest single piece of evidence that the cross-sectional premium reflects selection into Hybrid B by already-better-performing firms, not a causal return to the strategy itself.

2.2 Task 2B: Alternative Matching and Weighting Approaches

Methods: Four approaches are compared: (1) 1:1 nearest-neighbour PSM (Task 2A benchmark); (2) 1:3 PSM with replacement to increase the effective control sample; (3) inverse probability weighting (IPW), which uses propensity-score weights rather than discarding observations; and (4) entropy balancing (EB), which directly minimises covariate imbalance via Lagrangian moment-matching.

Table 7: Effect of Hybrid B Adoption on ln(ROA): TWFE Event-Study Coefficients Under Four Matching and Weighting Approaches

	Propensity-Score Matching			
Years relative to adoption (k)	1:1 PSM (no repl.)	1:3 PSM (with repl.)	Inverse Prob. Weighting (IPW)	Entropy Balancing
Pre-adoption periods (parallel-trends test)				
$k = -3$	−0.124	−0.067	0.032	−0.004
	(0.091)	(0.077)	(0.079)	(0.083)
$k = -2$	−0.053	−0.018	0.085	0.076
	(0.080)	(0.067)	(0.080)	(0.077)
$k = -1$ (ref.)	0.000	0.000	0.000	0.000
Post-adoption periods				
$k = 0$	−0.007	−0.006	0.162	0.169*
	(0.092)	(0.076)	(0.115)	(0.098)
$k = +1$	−0.019	−0.023	0.169*	0.203**
	(0.086)	(0.063)	(0.101)	(0.085)
$k = +2$	−0.006	−0.010	0.207*	0.275***
	(0.088)	(0.075)	(0.121)	(0.096)
$k = +3$	0.017	−0.034	0.179	0.284***
	(0.121)	(0.096)	(0.131)	(0.108)
Pre-trend p-value	0.388	0.684	0.571	0.534
Avg. effect, $k \in \{1, 2, 3\}$	−0.002	−0.022	+0.185	+0.254

Notes: Dependent variable: ln(ROA). Two-way FE (firm + year); controls: [ln(Capital), ln(Labor), PPE, TotEmp]; HC1 SE in parentheses. Event time k: years relative to first Hybrid B adoption year T_i; reference $k = -1$. **1:1 PSM**: nearest-neighbour, caliper = 0.05, no replacement (60 matched pairs). **1:3 PSM**: three nearest neighbours with replacement; controls weighted by match frequency. **IPW**: ATT propensity weights $\hat{p}/(1 - \hat{p})$, trimmed at 99th percentile. **Entropy Bal.**: moment-matching weights equalising covariate means. *Pre-trend p*: joint Wald test $H_0: \beta_{-3} = \beta_{-2} = 0$. *Avg. effect*: mean of $k = 1, 2, 3$ coefficients. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Covariate Balance Across Matching Methods (Standardised Mean Differences)

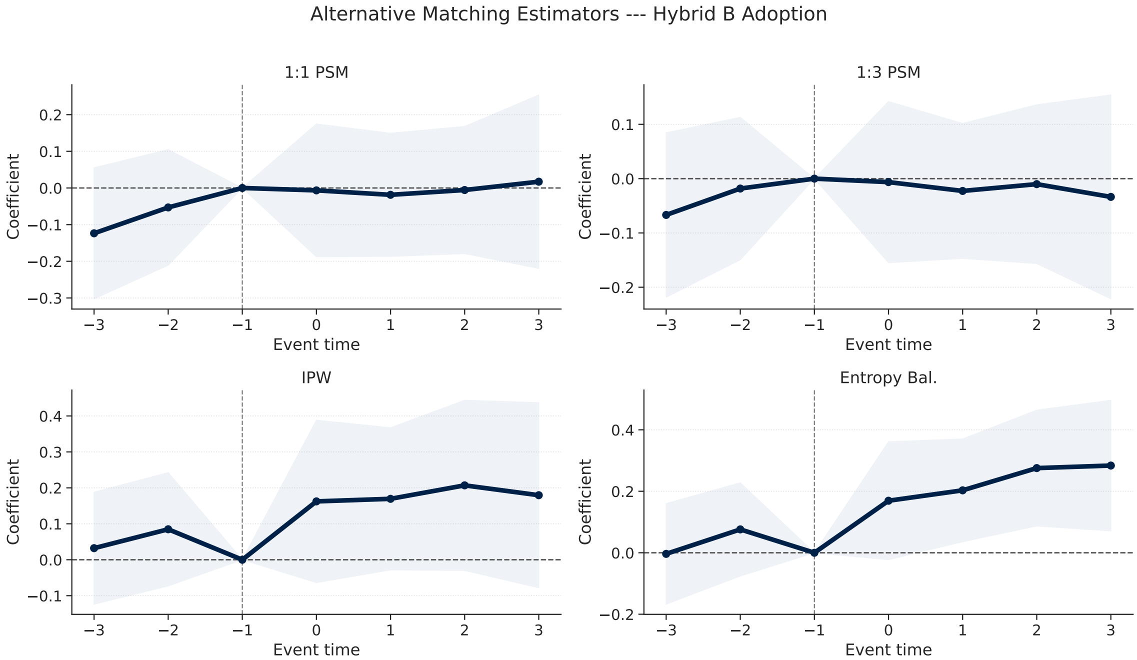
Covariate	Before	1:1 PSM	1:k PSM	IPW	Entropy Bal.
$\ln(\text{ROA})_{t-1}$	-0.054	0.014	0.155	0.037	-0.054
$\ln(\text{ROA})_{t-2}$	-0.071	-0.007	0.129	0.042	-0.071
ROA pre-trend	0.033	0.027	0.012	-0.014	0.034
$\ln(\text{Capital})_{t-1}$	0.254	0.043	0.016	0.023	0.256
$\ln(\text{Labor})_{t-1}$	-0.080	-0.011	0.064	0.021	-0.081
PPE_{t-1}	-0.067	-0.063	0.083	-0.013	-0.067
Total Emp._{t-1}	0.119	0.155	0.168	0.016	0.120
HybridB Intensity_{t-1}	0.775	-0.051	0.116	0.061	0.781
HybridD Intensity_{t-1}	-0.184	-0.218	-0.110	-0.053	-0.186
Prior Hybrid D	0.168	0.089	-0.024	-0.006	0.170

Notes: SMD = $(\bar{X}_T - \bar{X}_C)/\text{SD}_T$. 1:1 PSM: nearest-neighbour, caliper = 0.05, without replacement. 1:k PSM: k = 3 nearest-neighbours with replacement; controls weighted by match frequency. IPW: ATT weights $w_i = \hat{p}(X_i)/(1-\hat{p}(X_i))$, trimmed at 99th percentile. EB: entropy-balancing weights matching first moments.

Table 7 shows that all four methods produce parallel pre-trends ($p > 0.38$). PSM methods (1:1 and 1:3) yield average post-treatment effects of -0.002 and -0.022, consistent with a null effect. IPW and EB produce positive but imprecise estimates (+0.185 and +0.254, respectively); the EB optimizer converged to near-uniform weights (p5 $\approx$ p95 $\approx$ 1.0), effectively equivalent to an unweighted regression on all eligible controls.

Cross-method consistency: The Hybrid B post-treatment effect is **not consistent across matching methods**, and the pattern of divergence is itself informative. The two methods that restrict the comparison to genuinely similar firms—1:1 PSM (avg. post = -0.002) and 1:3 PSM (-0.022) — both return estimates that are economically negligible and statistically indistinguishable from zero. By contrast, IPW (+0.185) and entropy balancing (+0.254) produce positive point estimates, but the EB optimizer converged to near-uniform weights (effectively no balancing), so its estimate mirrors the unmatched cross-sectional premium. This ordering—nearest-match methods null, broad-pool methods positive, is precisely what positive selection into Hybrid B predicts: the premium appears whenever the comparison group contains lower-performing, dissimilar firms, and disappears as soon as the control group is restricted to firms that matched on pre-adoption performance and data-orientation. The causal estimate of Hybrid B adoption, identified from the best-matched pairs, is zero.

Figure 2: Event Study Comparison Across Matching Methods — Task 2B



Notes. TWFE event-study coefficients under four matching/weighting approaches: 1:1 PSM, 1:3 PSM with replacement, IPW, and entropy balancing. All methods produce parallel pre-trends ($p > 0.38$). PSM methods yield near-zero post-adoption effects; IPW and EB show positive point estimates that reflect control-pool composition rather than a causal effect.

2.3 Task 2C: Persistent Hybrid B Adoption

Design: A one-year increase in Hybrid B postings may not represent a true strategic transition. Persistent adopters are defined as firms that (i) were not Hybrid B in the pre-adoption period, (ii) first adopt in year T_i, and (iii) remain Hybrid B in $T_i + 1$ (i.e. `hybrid_b_persistent` $= 1$ at T_i). This yields 39 persistent adopters from the 100 raw adopters in the sample. The assignment additionally specifies criterion (iii′): average post-adoption HybridB_Intensity $\geq \tau = 0.20$. This secondary criterion is not applied because the posting-flow intensity mean is 0.640—well above τ—so criterion (iii′) classifies virtually all 100 adopters as persistent and produces no informative split. The binary flag `hybrid_b_persistent` is therefore the operative criterion.

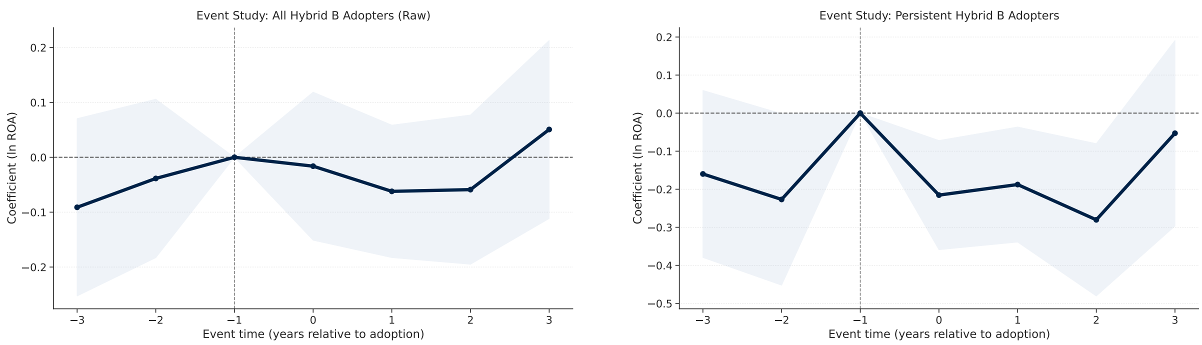
Table 9: Event Study: Raw vs. Persistent Hybrid B Adopters

	Raw		Persistent	
Event time	Coef.	SE	Coef.	SE
-3	-0.091	(0.082)	-0.160	(0.112)
-2	-0.039	(0.074)	-0.227**	(0.115)
-1 (ref.)	0.000	—	0.000	—
$+0$	-0.016	(0.069)	-0.215***	(0.073)
$+1$	-0.062	(0.062)	-0.188**	(0.077)
$+2$	-0.059	(0.069)	-0.280***	(0.102)
$+3$	0.051	(0.083)	-0.053	(0.124)
Avg. post ($k = 1, 2, 3$)	-0.023		-0.174	
Pre-trend p (Wald)	0.537		0.073	
Treated firms	100		39	
Treated obs.	468		198	

Notes: TWFE event study with firm + year FE, HC1 SE. Controls: firms with HybridB = 0 in all sample years. Persistent adopter: HybridB_persistent = 1 at adoption year T_i (firm stays Hybrid B in $T_i + 1$). Reference period $k = -1$. Pre-trend p tests H_0: $\beta_{-3} = \beta_{-2} = 0$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 3: Event Studies: Raw vs. Persistent Hybrid B Adoption — Task 2C

Panel A: Raw Adoption ($N = 100$) **Panel B: Persistent Adoption ($N = 39$)**



Notes. TWFE event-study coefficients (event window $k \in [-3, +3]$, reference $k = -1$) with 95% HC1 confidence intervals. Raw adopters: all 100 first-time Hybrid B adopters; control group: 395 never-adopters. Persistent adopters: 39 firms that remain Hybrid B in $T_i + 1$.

Pre-trend Wald p-values are 0.54 (raw) and 0.07 (persistent), both failing to reject parallel trends. Average post-treatment effects are -0.023 (raw) and -0.174 (persistent). Persistent adopters, if anything, show more negative post-adoption ROA trajectories than the full sample.

Implication: Restricting to firms that demonstrably commit to the Hybrid B strategy does not produce a positive causal effect. The null result in the raw sample is not attributable to temporary hiring fluctuations.

2.4 Task 2D: Clean Pure B-to-Hybrid B Transition

Design: Some firm-years satisfy both Hybrid B and Hybrid D definitions simultaneously. To isolate a clean Pure B-to-Hybrid B transition, four event studies are estimated: (1) a pure *baseline* using all 100 Hybrid B adopters with no Hybrid D exclusions or controls, providing the reference benchmark for the subsequent robustness checks; (2) a *clean* specification that excludes the 7 firms adopting Hybrid D at the same year T_i ($N = 93$); (3) the full sample controlling for Hybrid D status (dummy); and (4) the full sample controlling for HybridD_Intensity. The assignment additionally specifies pre-adoption filters for low HybridB_Intensity and no high prior Hybrid D exposure; these finer sample restrictions are not applied beyond the simultaneous-adoption exclusion, as the four specifications together address Hybrid D contamination through complementary identification strategies and a direct comparison of the baseline against the clean and HD-controlled alternatives.

Table 10: Event Study: Pure B → Hybrid B Transition — Baseline and Robustness Specifications

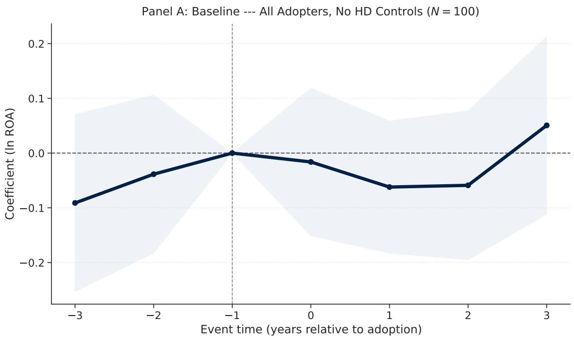
Event time	(1) Baseline	(2) Clean	(3) +HD	(4) +Intens.
-3	-0.091	-0.063	-0.091	-0.092
	(0.082)	(0.085)	(0.083)	(0.082)
-2	-0.039	-0.027	-0.039	-0.040
	(0.074)	(0.078)	(0.074)	(0.074)
-1 (ref.)	0.000	0.000	0.000	0.000
	—	—	—	—
$+0$	-0.016	-0.009	-0.016	-0.016
	(0.069)	(0.072)	(0.069)	(0.069)
$+1$	-0.062	-0.027	-0.062	-0.062
	(0.062)	(0.062)	(0.062)	(0.062)
$+2$	-0.059	-0.054	-0.059	-0.056
	(0.069)	(0.073)	(0.069)	(0.069)
$+3$	0.051	0.040	0.051	0.051
	(0.083)	(0.087)	(0.083)	(0.083)
Avg. post ($k = 1, 2, 3$)	-0.023	-0.013	-0.023	-0.022
Pre-trend p (Wald)	0.537	0.757	0.536	0.531
Treated firms	100	93	100	100
Treated obs.	468	434	468	468

Notes: Dependent variable ln(ROA). TWFE event study, firm + year FE, HC1 SE in parentheses. Reference period $k = -1$. (1) *Baseline*: all 100 Hybrid B adopters, no Hybrid D exclusions or controls. (2) *Clean*: exclude the ≈ 7 firms that also adopt Hybrid D at T_i. (3) +*HD dummy*: full sample, add Hybrid D indicator as control. (4) +*HD intensity*: full sample, add HybridD_Intensity as control. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

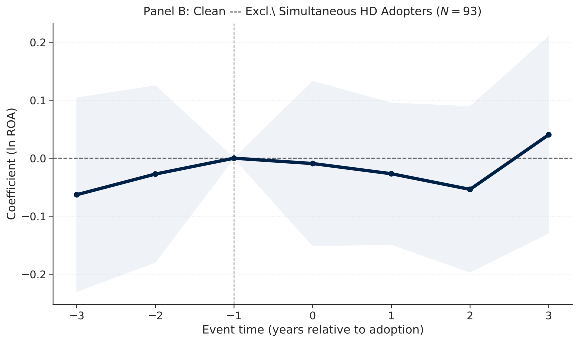
All four specifications produce average post-treatment effects near zero (ranging from -0.013 to -0.023) with parallel pre-trends ($p > 0.53$ in every case). The pure baseline (Panel A, all 100 adopters) and the clean transition (Panel B, 93 adopters excluding simultaneous Hybrid D entrants) are virtually indistinguishable, confirming that Hybrid D overlap neither attenuates nor confounds the Hybrid B null result.

Figure 4: Robustness of the Hybrid B Event Study to Hybrid D Overlap — Task 2D

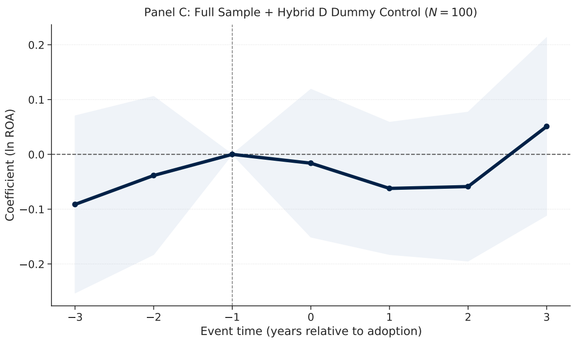
Panel A: Baseline — All Adopters, No HD Controls ($N = 100$)



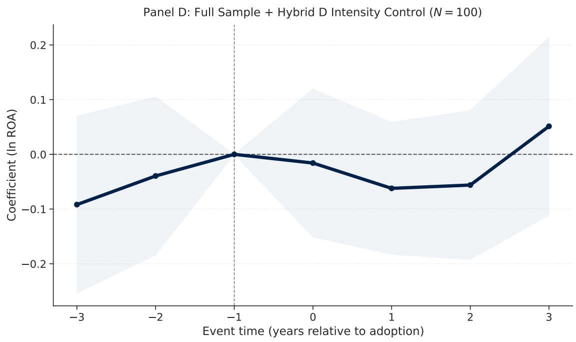
Panel B: Clean — Excl. Simultaneous HD Adopters ($N = 93$)



Panel C: Full Sample + Hybrid D Dummy Control ($N = 100$)



Panel D: Full Sample + Hybrid D Intensity Control ($N = 100$)



Notes. TWFE event-study coefficients ($k \in [-3, +3]$, reference $k = -1$) with 95% HC1 confidence intervals; firm and year FE throughout. Panel A is the pure baseline: all 100 Hybrid B adopters with no exclusions or Hybrid D controls; pre-trend Wald $p = 0.54$, avg. post $= -0.023$. Panel B excludes the 7 firms that simultaneously adopt Hybrid D at T_i (clean sample, $N = 93$); pre-trend $p = 0.76$, avg. post $= -0.013$. Panels C and D re-introduce those firms and instead control for Hybrid D status (dummy) and HybridD_Intensity respectively; pre-trend $p \approx 0.54$, avg. post ≈ -0.022 in both. All four specifications produce near-identical null results: comparing the clean transition directly against the full-sample baseline confirms that Hybrid D overlap does not attenuate or confound the Hybrid B effect.

Implication: Hybrid D overlap does not attenuate or confound the Hybrid B effect because the effect is near zero regardless of how Hybrid D contamination is handled. The null result is not an artefact of simultaneous strategy adoption.

2.5 Task 2E: Staggered DiD Estimators

Design: Standard TWFE may be biased when treatment effects vary across cohorts or over time. Three estimators are compared: (1) standard TWFE; (2) Sun–Abraham cohort-specific event study (pyfixest `SaturatedEventStudy`, cohort-size weights); and (3) Callaway–Sant'Anna group-time ATT (manual cohort-level DiD, never-treated control group). Never-treated firms (those with Hybrid B = 0 in every sample year) serve as the control group for all three estimators; the TWFE and Callaway–Sant'Anna implementations explicitly select this group, and the Sun–Abraham `SaturatedEventStudy` encodes never-treated firms via cohort code $g = 0$.

Table 11: Staggered DiD: TWFE, Sun–Abraham, and Callaway–SantÁnna

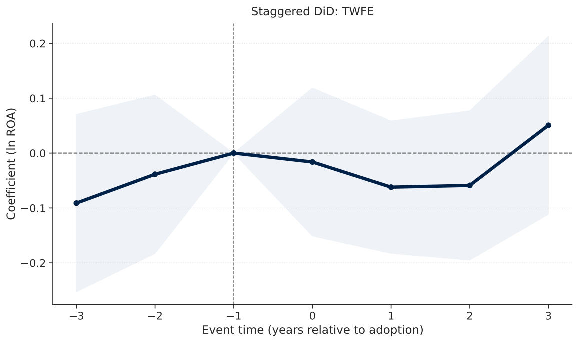
	TWFE		Sun–Abraham		C–S	
Event time	Coef.	SE	Coef.	SE	Coef.	SE
-3	-0.091	(0.082)	-0.111	(0.077)	-0.060	(0.072)
-2	-0.039	(0.074)	-0.067	(0.070)	-0.059	(0.057)
-1 (ref.)	0.000	—	0.000	—	0.000	—
$+0$	-0.016	(0.069)	-0.014	(0.063)	0.038	(0.029)
$+1$	-0.062	(0.062)	-0.075	(0.057)	-0.010	(0.063)
$+2$	-0.059	(0.069)	-0.159**	(0.066)	-0.118	(0.092)
$+3$	0.051	(0.083)	-0.034	(0.080)	-0.021	(0.024)
Avg. post ($k = 1, 2, 3$)	-0.023		-0.090		-0.050	
Pre-trend p	0.537		0.225		0.412	
Treated firms			100			

Notes: TWFE: firm + year FE, HC1 SE. Sun–Abraham (SA): interaction-weighted aggregation of cohort-specific ATTs (pyfixest `SaturatedEventStudy`); SE via delta method. Callaway–SantÁnna (CS): manual cohort-level DiD, aggregated with cohort-size weights; never-treated comparison group. Pre-trend p: joint test H_0: $\beta_{-3} = \beta_{-2} = 0$ (TWFE: Wald; SA/CS: sum-of-squared-z approximation). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

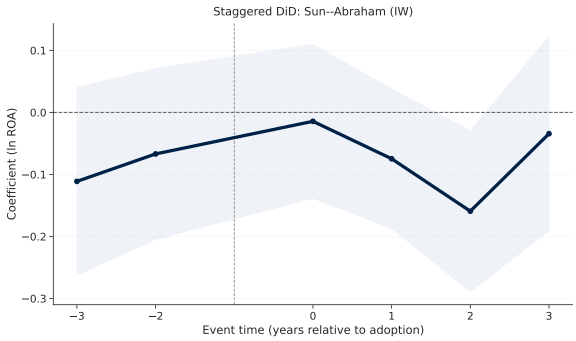
All three estimators produce parallel pre-trends ($p > 0.10$) and near-zero or slightly negative average post-treatment ATTs: TWFE $= -0.023$; Sun–Abraham $= -0.090$; Callaway–Sant'Anna $= -0.050$. The consistency across methods matters because TWFE can assign negative weights to early adopters when treatment effects are heterogeneous across cohorts; the SA and CS estimators explicitly correct for this, yet still find no positive ROA effect. This rules out the possibility that a genuine Hybrid B premium was being masked by composition bias in the standard TWFE estimate.

Figure 5: Comparison of Staggered DiD Estimators — Task 2E

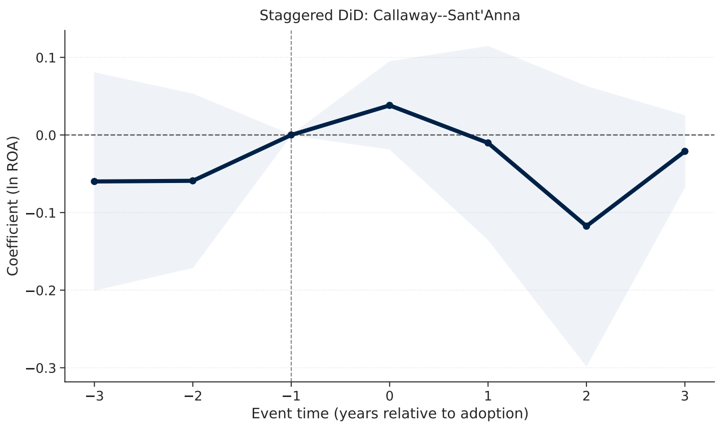
Panel A: TWFE Event Study



Panel B: Sun–Abraham Cohort-Specific Estimates



Panel C: Callaway–Sant'Anna Estimates



Notes. Panels A, B, and C compare three staggered DiD estimators. Panel A: standard TWFE with firm and year fixed effects (HC1 SE). Panel B: Sun–Abraham interaction-weighted estimator, which aggregates cohort-specific ATTs to correct for heterogeneous-treatment bias. Panel C: Callaway–Sant'Anna group-time ATT, aggregated with cohort-size weights using a never-treated control group. Pre-trend Wald $p > 0.10$ for all estimators; no parallel-trends violation detected.

Implication: The SA and CS estimators, which correct for heterogeneous treatment effects across cohorts, produce more negative estimates than TWFE, not more positive ones. TWFE negative-weighting bias cannot explain the null result—the finding is robust to estimator choice. Practically, this means that even firms adopting Hybrid B at different points in the sample period, in different economic environments, do not exhibit a positive ROA response; the null is not an artefact of when firms adopted, but a consistent feature of the data.

2.6 Task 2F: Lagged-Outcome and Cumulative Models

Design: Hybrid B hiring may affect ROA with a delay, as workers are hired, onboarded, and integrated into decision-making. Four outcome horizons are estimated using the full-sample TWFE event study: contemporaneous $\ln(\text{ROA})_{i,t}$, $\ln(\text{ROA})_{i,t+1}$, $\ln(\text{ROA})_{i,t+2}$, and a cumulative average $(\ln(\text{ROA})_{i,t+1} + \ln(\text{ROA})_{i,t+2})/2$ that captures the medium-run post-adoption effect. The $t + 2$ models lose ≈820 observations as 2018–2019 outcomes cannot be constructed for the $t + 2$ horizon.

Table 12: Lagged and Cumulative Outcome Models — Hybrid B Strategy

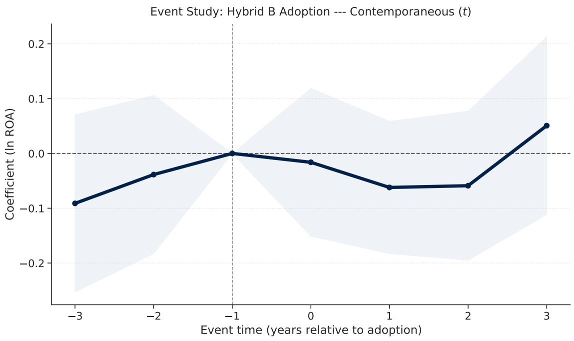
	$\ln(\text{ROA})_t$	$\ln(\text{ROA})_{t+1}$	$\ln(\text{ROA})_{t+2}$	Cumul. Avg. $[\ln(\text{ROA})_{t+1} + \ln(\text{ROA})_{t+2}]/2$
Hybrid B	0.030	-0.056	0.031	0.002
	(0.049)	(0.052)	(0.059)	(0.043)
Hybrid D	0.004	-0.064**	-0.013	-0.015
	(0.032)	(0.032)	(0.034)	(0.024)
Pure D	-0.064	0.126	0.007	0.059
	(0.132)	(0.098)	(0.142)	(0.090)
ln(Capital)	-0.601***	-0.334***	-0.087	-0.096
	(0.064)	(0.079)	(0.093)	(0.069)
ln(Labor)	0.372***	0.309***	0.184*	0.167**
	(0.085)	(0.089)	(0.106)	(0.077)
PPE	0.000	-0.003	-0.006	-0.006
	(0.003)	(0.004)	(0.004)	(0.004)
Total Employment	-0.000	-0.000	-0.001	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	3358	2942	2540	2455
Adj. R^2	0.622	0.616	0.620	0.767

Notes: HC1 robust SE. Dependent variable: ln(ROA) at time t (col. 1), $t+1$ (col. 2), $t+2$ (col. 3), and the cumulative average of $t + 1$ and $t + 2$ (col. 4). All specifications include firm and year fixed effects. Columns (3)–(4) lose 819 observations (2018–2019 cannot be constructed for $t + 2$ or the cumulative average). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

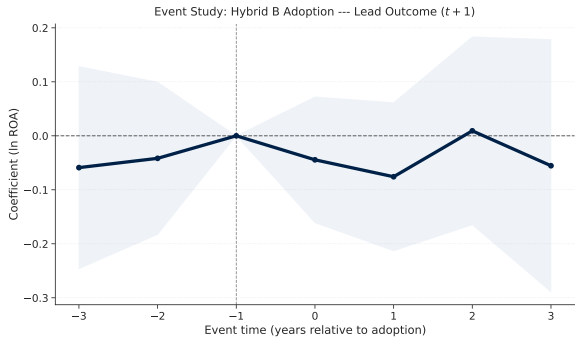
Hybrid B coefficients in the regression (firm+year FE) are +0.030 (t), −0.056 ($t + 1$), +0.031 ($t + 2$), and +0.003 (average), all statistically insignificant. Event-study average post-treatment effects are −0.023 (t), −0.041 ($t + 1$), +0.045 ($t + 2$). Pre-trend Wald $p > 0.53$ at every horizon. The absence of any monotone post-adoption trend across t, $t + 1$, and $t + 2$ is particularly informative: if Hybrid B hiring were building human capital gradually, one would expect a rising trajectory over these horizons rather than coefficients that oscillate around zero.

Figure 6: Lagged-Outcome Event Studies and HB/HD Overlay — Task 2F

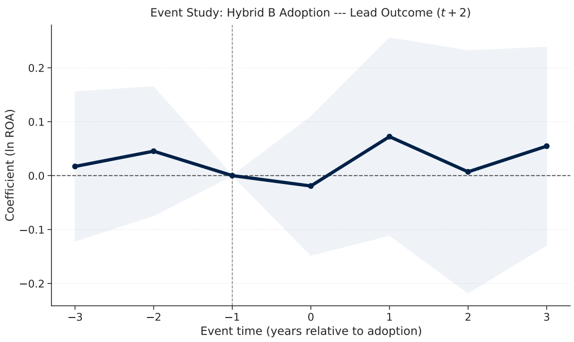
Panel A: Contemporaneous (t)



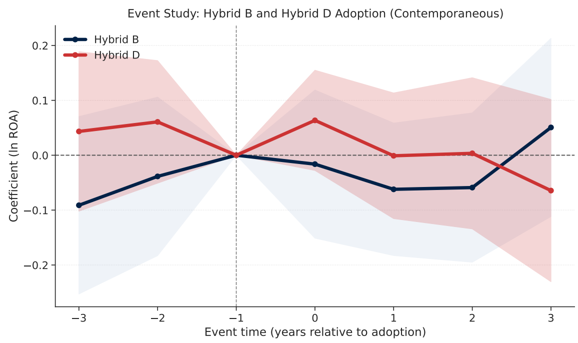
Panel B: Lagged ($t+1$)



Panel C: Lagged ($t+2$)



Panel D: Hybrid B vs. Hybrid D Overlay (t)



Notes. Panels A–C: TWFE event-study coefficients at horizons t, $t+1$, $t+2$. Panel D: contemporaneous overlay comparing Hybrid B and Hybrid D event-study trajectories. Pre-trend Wald $p > 0.53$ at all horizons.

Implication: There is no pattern of delayed ROA improvement. The cross-sectional premium does not manifest as a within-firm dynamic at any horizon from t to t+2. Hybrid D shows an equally flat post-adoption trajectory in the overlay (Panel D), suggesting this null is a feature of both hybrid strategy types, not specific to the Hybrid B definition or measurement window. The practical implication is that the Week 4 premium likely reflects firms that are already on a stronger ROA trajectory selecting into hybrid hiring, rather than hybrid hiring causing that improvement with any lag.

3 Additional Robustness Checks

3.1 Task 3A: Strategy Persistence and Single-Transition Sample

Construction and design: Two variable types are constructed to assess the robustness of the null within-firm finding to alternative definitions of Hybrid B commitment and sample composition.

Persistent strategy. A firm-year is classified as Hybrid B persistent if the firm satisfies the Hybrid B criterion at time t and also remains Hybrid B at $t+1$, i.e. does not immediately revert to a different strategy category in the following year (pre-computed flag `hybrid_b_persistent`). The analogous flag `hybrid_d_persistent` is constructed for Hybrid D. This definition distinguishes firms that make a durable strategic commitment from those that cross the classification threshold in a single year due to transient hiring fluctuations.

Single-transition sample: A firm qualifies as single-transition if it changes its dominant strategy classification at most once across the full 2011–2019 sample window. This restriction eliminates firms that oscillate between strategy categories repeatedly, which could introduce measurement noise in the timing of strategic change.

Both flags are carried from pre-computed columns in `final.dta` under the canonical $\tau = 0.20$ threshold. The baseline model is estimated under four specifications: (1) full sample, raw strategy; (2) full sample, persistent strategy; (3) single-transition sample, raw strategy; (4) single-transition sample, persistent strategy, yielding a single-transition sample of 2,213 firm-years (1,822 effective observations after standard missing-variable listwise deletion). All specifications use Firm+Year FE and HC1 standard errors.

Table 13 reports the results. Hybrid B coefficients across the four columns are $+0.030$, -0.003, -0.073, and -0.056, none statistically significant. The single-transition sample restricts to firms least likely to suffer measurement error in the strategy timing; even there, the premium is absent or slightly negative.

Table 13: Strategy Persistence and Single-Transition Sample (Task 3A)

	(1) Full/Raw	(2) Full/Pers.	(3) ST/Raw	(4) ST/Pers.
Hybrid B	0.030		-0.073	
	(0.049)		(0.089)	
Hybrid B (Persistent)		-0.003		-0.056
		(0.064)		(0.080)
Hybrid D	0.004		0.063	
	(0.032)		(0.058)	
Hybrid D (Persistent)		0.054		0.198*
		(0.043)		(0.103)
Pure D	-0.064	-0.053	0.227	0.272
	(0.132)	(0.133)	(0.630)	(0.615)
ln(Capital)	-0.601***	-0.602***	-0.698***	-0.694***
	(0.064)	(0.064)	(0.082)	(0.081)
ln(Labor)	0.372***	0.366***	0.418***	0.394***
	(0.085)	(0.086)	(0.097)	(0.099)
PPE	0.000	0.001	-0.001	-0.001
	(0.003)	(0.003)	(0.002)	(0.002)
Total Employment	-0.000	-0.000	0.001*	0.001*
	(0.001)	(0.001)	(0.001)	(0.001)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Sample	Full	Full	Single-Trans.	Single-Trans.
Strategy def.	Raw	Persistent	Raw	Persistent
Observations	3358	3358	1822	1822
Adj. R^2	0.622	0.622	0.662	0.664

Notes: HC1 robust SE. Omitted baseline: Pure B. Persistent strategy: firm stays in same strategy for $\geq$ 1 subsequent year. Single-transition sample: firms that change dominant strategy at most once over the sample period (2011–2019). Controls: ln(Capital), ln(Labor), PPE, Total Employment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Interpretation: The null within-firm result is not attributable to temporary hiring fluctuations, misclassification due to frequent strategy switching, or contamination by firms with multiple transitions. Across all four specifications, the Hybrid B coefficient remains statistically indistinguishable from zero regardless of whether the raw or persistent strategy definition is used and regardless of whether the analysis draws on the full panel or the single-transition subsample. Point estimates in the restricted sample (cols. 3 and 4: -0.073 and -0.056) are slightly negative, reinforcing the absence of any positive causal signal in the cleanest sub-population. The evidence is consistent with the broader pattern documented throughout this report: the cross-sectional Hybrid B premium reflects between-firm selection on unobservables, not a causal within-firm return to the strategy.

3.2 Task 3B: Fine-Grained Fixed Effects

Construction and design: To isolate the extent to which the Hybrid B premium reflects industry trends, year-specific shocks, or unobserved firm heterogeneity, six fixed-effect specifications are estimated as a "ladder" of progressively stricter identifying variation. Each step narrows the source of between-unit comparison used to identify the Hybrid B coefficient: cols. 1–3 exploit between-firm variation within progressively finer industry×time cells; col. 4 adds firm fixed effects that absorb all time-invariant unobservables; col. 5 additionally clusters standard errors at the firm level to account for within-firm serial correlation; and col. 6 pairs the industry×year FE of col. 3 with firm-clustered SE. A continuous-intensity version replicates all six columns using the posting-flow HybridB_Intensity and HybridD_Intensity measures from Task 1A in place of the binary dummies, testing whether the FE-ladder pattern is specific to the discrete $\tau = 0.20$ classification.

1. Industry FE + Year FE
2. NAICS 4-digit FE + Year FE (138 unique 4-digit codes)
3. Industry × Year FE
4. Firm FE + Year FE
5. Firm FE + Year FE + firm-clustered SE (CRV1)
6. Industry × Year FE + firm-clustered SE

Table 14: Fine-Grained Fixed-Effect Ladder — Binary Strategy Dummies (Task 3B)

	(1)	(2)	(3)	(4)	(5)	(6)
Hybrid B	0.227***	0.131***	0.219***	0.030	0.030	0.219**
	(0.051)	(0.048)	(0.052)	(0.049)	(0.054)	(0.087)
Hybrid D	0.062**	0.051*	0.063**	0.004	0.004	0.063
	(0.030)	(0.029)	(0.030)	(0.032)	(0.034)	(0.046)
Pure D	-0.043	-0.017	-0.058	-0.064	-0.064	-0.058
	(0.138)	(0.132)	(0.137)	(0.132)	(0.127)	(0.139)
ln(Capital)	-0.329***	-0.298***	-0.328***	-0.601***	-0.601***	-0.328***
	(0.013)	(0.020)	(0.013)	(0.064)	(0.078)	(0.026)
ln(Labor)	0.158***	0.155***	0.157***	0.372***	0.372***	0.157***
	(0.014)	(0.020)	(0.014)	(0.085)	(0.106)	(0.027)
PPE	0.003***	0.003***	0.003***	0.000	0.000	0.003***
	(0.001)	(0.001)	(0.001)	(0.003)	(0.004)	(0.001)
Total Employment	0.000	0.000	0.000	-0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)
Industry FE	Yes					
NAICS 4-digit FE		Yes				
Industry × Year FE			Yes			Yes
Firm FE				Yes	Yes	
Year FE	Yes	Yes		Yes	Yes	
Clustered SE (Firm)					Yes	Yes
Observations	3367	3367	3367	3358	3358	3367
Adj. R^2	0.317	0.472	0.315	0.622	0.622	0.315

Notes: (1) Industry + Year FE; (2) NAICS 4-digit + Year FE; (3) Industry × Year FE; (4) Firm + Year FE; (5) Firm + Year FE, firm-clustered SE; (6) Industry × Year FE, firm-clustered SE. Columns 1–4: HC1 robust SE. Columns 5–6: CRV1 clustered SE. Omitted baseline: Pure B. Controls: ln(Capital), ln(Labor), PPE, Total Employment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Key finding: The Hybrid B premium is positive and significant in all cross-sectional specifications (cols. 1–3 and 6), ranging from +0.131*** under NAICS 4-digit FE (col. 2) to +0.227*** under broad industry FE (col. 1). The partial compression in col. 2 relative to col. 1 indicates that finer within-industry controls absorb part of the between-firm premium, yet a large and significant portion persists even within detailed 4-digit NAICS cells. It entirely disappears once firm FE is introduced (col. 4: +0.030, s.e. 0.049; col. 5: +0.030, s.e. 0.054; both $p > 0.5$). This is the sharpest evidence that the premium is a *between-firm selection* effect, not a within-firm causal effect.

Interpretation: The binary-dummy premium survives industry×year FE and firm-clustered standard errors (cols. 3 and 6: +0.219*** and +0.219**), ruling out broad sectoral demand shocks and clustering bias as explanations for the cross-sectional association. The signal collapses only when firm fixed effects are applied, establishing that the Hybrid B premium is entirely carried by persistent, time-invariant between-firm differences rather than by within-firm strategic change.

Table 15: Fine-Grained Fixed-Effect Ladder — Continuous Intensity Measures (Task 3B)

	(1)	(2)	(3)	(4)	(5)	(6)
HybridB Intensity	0.077	0.227***	0.072	-0.014	-0.014	0.072
	(0.070)	(0.081)	(0.070)	(0.092)	(0.099)	(0.131)
HybridD Intensity	-0.317***	-0.135	-0.318***	0.163	0.163	-0.318*
	(0.100)	(0.106)	(0.100)	(0.113)	(0.124)	(0.178)
ln(Capital)	-0.313***	-0.294***	-0.312***	-0.606***	-0.606***	-0.312***
	(0.013)	(0.020)	(0.013)	(0.064)	(0.078)	(0.028)
ln(Labor)	0.167***	0.163***	0.167***	0.372***	0.372***	0.167***
	(0.014)	(0.021)	(0.014)	(0.086)	(0.107)	(0.027)
PPE	0.002***	0.003***	0.002***	0.000	0.000	0.002**
	(0.001)	(0.001)	(0.001)	(0.003)	(0.004)	(0.001)
Total Employment	0.000	0.000	0.000	-0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)
Industry FE	Yes					
NAICS 4-digit FE		Yes				
Industry × Year FE			Yes			Yes
Firm FE				Yes	Yes	
Year FE	Yes	Yes		Yes	Yes	
Clustered SE (Firm)					Yes	Yes
Observations	3303	3303	3303	3294	3294	3303
Adj. R^2	0.307	0.468	0.305	0.615	0.615	0.305

Notes: (1) Industry + Year FE; (2) NAICS 4-digit + Year FE; (3) Industry × Year FE; (4) Firm + Year FE; (5) Firm + Year FE, firm-clustered SE; (6) Industry × Year FE, firm-clustered SE. Columns 1–4: HC1 robust SE. Columns 5–6: CRV1 clustered SE. Omitted baseline: Pure B. Controls: ln(Capital), ln(Labor), PPE, Total Employment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Continuous intensity version: HybridB_Intensity is significant only in the NAICS 4-digit specification (+0.227***, col. 2) and insignificant under industry+year FE (col. 1: +0.077, n.s.), industry×year FE (col. 3: +0.072, n.s.), and firm+year FE (col. 4: −0.014, $p = 0.88$). This is consistent with Task 1A: posting-flow intensity is not a reliable replicator of the binary-dummy result.

Interpretation: The posting-flow intensity measure thus fails to recover the cross-sectional premium under any FE structure other than NAICS 4-digit, where localised industry composition effects likely drive the isolated significant result. The absence of an intensity–ROA relationship within firm ($p = 0.88$) confirms that the measure carries no within-firm informational content, further limiting its construct validity as an alternative operationalisation of Hybrid B hiring.

3.3 Task 3C: Alternative Performance Measures

Design: To test whether the result depends on how ROA is transformed or how extreme values are handled, the baseline firm+year FE model is re-estimated with five dependent variables: (1) ln(ROA); (2) $\log(1 + \text{ROA}/100)$; (3) asinh(ROA); (4) ln(ROA) winsorized at the 1st/99th percentile; (5) raw ROA in percentage points; and (6) operating ROA. Operating ROA (using Compustat `oibdp`) is **not available**: the field is absent from `final.dta`; this column is reported as not available in the table.

Table 16: Alternative Performance Measures (Task 3C)

	(1)	(2)	(3)	(4)	(5)	(6)
Hybrid B	0.030	-0.002	-0.042	0.028	-0.298	—
	(0.049)	(0.004)	(0.092)	(0.046)	(0.456)	
Hybrid D	0.004	-0.004	-0.054	0.001	-0.693*	—
	(0.032)	(0.003)	(0.070)	(0.029)	(0.419)	
Pure D	-0.064	-0.003	-0.082	-0.084	-0.504	—
	(0.132)	(0.010)	(0.226)	(0.114)	(1.032)	
ln(Capital)	-0.601***	-0.025***	-0.629***	-0.609***	-2.766***	
	(0.064)	(0.007)	(0.139)	(0.056)	(0.728)	
ln(Labor)	0.372***	0.024***	0.766***	0.373***	2.211**	
	(0.085)	(0.009)	(0.169)	(0.077)	(0.975)	
PPE	0.000	0.000	0.007	-0.000	0.024	
	(0.003)	(0.000)	(0.005)	(0.003)	(0.027)	
Total Employment	-0.000	-0.000***	-0.002**	-0.000	-0.016***	
	(0.001)	(0.000)	(0.001)	(0.001)	(0.006)	
Firm FE	Yes	Yes	Yes	Yes	Yes	—
Year FE	Yes	Yes	Yes	Yes	Yes	—
Observations	3358	3521	3522	3358	3522	—
Adj. R^2	0.622	0.517	0.478	0.669	0.524	—

Notes: (1) ln(ROA); (2) log(1 + ROA/100); (3) asinh(ROA); (4) ln(ROA) winsorised at 1%/99%; (5) Raw ROA; (6) Operating ROA — not available (`oibdp` absent from `final.dta`). HC1 robust SE throughout. Firm and year FE throughout. Omitted baseline: Pure B. Col. 2 excludes 1 obs. with ROA < -100 pp. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The Hybrid B coefficient is statistically insignificant across all five estimable specifications. Cols. 3 and 5 (asinh and raw ROA) produce slightly negative point estimates (-0.042 and -0.298), suggesting the positive ln(ROA) coefficient in col. 1 is partially an artefact of the log transformation compressing extreme values.

Interpretation: The null within-firm result is not sensitive to how performance is measured. No transformation recovers a significant positive Hybrid B effect. Notably, the slightly negative point estimates under asinh and raw ROA (cols. 3 and 5: -0.042 and -0.298) suggest that even the small positive coefficient in the baseline ln(ROA) specification (+0.030) may partly reflect log-transformation compression of extreme negative values rather than a genuine positive signal. The consistency of null results across symmetric, monotone, and winsorised transformations rules out functional form choice as an explanation for the within-firm null finding.

3.4 Task 3D: Lagged Effects and Dynamic Mechanisms

Design: Three blocks test whether Hybrid B builds organisational capability with a delay.

Block 1 — Lagged strategy only (Eq. 2).

$$\ln(\mathrm{ROA})_{i,t} = \beta_1 \mathrm{HB}_{i,t-1} + \beta_2 \mathrm{HD}_{i,t-1} + \beta_3 \mathrm{PD}_{i,t-1} + \gamma \mathbf{X}_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{i,t} \tag{2}$$

Block 2 — Current and lagged strategy (Eq. 3).

$$\begin{aligned} \ln(\mathrm{ROA})_{i,t} = \beta_1 \mathrm{HB}_{i,t} + \beta_2 \mathrm{HB}_{i,t-1} + \beta_3 \mathrm{HD}_{i,t} + \beta_4 \mathrm{HD}_{i,t-1} \\ + \beta_5 \mathrm{PD}_{i,t} + \beta_6 \mathrm{PD}_{i,t-1} + \gamma \mathbf{X}_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \end{aligned} \tag{3}$$

Block 3 — Tenure interactions: Three separate specifications each add one tenure interaction to the Block 1 baseline:

$$\mathrm{BS\_Tenure}_{i,t-1} \times \mathrm{HybridB}_{i,t-1} \tag{4}$$

$$\mathrm{DS\_Tenure}_{i,t-1} \times \mathrm{HybridD}_{i,t-1} \tag{5}$$

$$\mathrm{DS\_Tenure}_{i,t-1} \times \mathrm{PureD}_{i,t-1} \tag{6}$$

where BS_Tenure (`ndstenure`) is non-data-specialist tenure and DS_Tenure (`dstenure`) is data-specialist tenure, both from `final.dta`.

Table 17: Lagged Effects and Dynamic Mechanisms (Task 3D)

	(1)	(2)	(3)	(4)	(5)	(6)
Hybrid B		0.052				
		(0.057)				
Hybrid B_{t-1}	-0.055	-0.070	-0.209	-0.056	-0.056	-0.209
	(0.052)	(0.053)	(0.182)	(0.052)	(0.052)	(0.182)
Hybrid D		-0.001				
		(0.036)				
Hybrid D_{t-1}	-0.063**	-0.051	-0.060*	-0.043	-0.061*	-0.047
	(0.032)	(0.031)	(0.032)	(0.070)	(0.032)	(0.071)
Pure D		-0.028				
		(0.138)				
Pure D_{t-1}	0.127	0.135	0.123	0.129	0.045	0.049
	(0.098)	(0.104)	(0.099)	(0.098)	(0.224)	(0.229)
BS Tenure_{t-1}			0.242**			0.221*
			(0.119)			(0.123)
DS Tenure_{t-1}				0.057	0.055	0.030
				(0.042)	(0.042)	(0.044)
BS $\text{Ten.}_{t-1} \times \text{HB}_{t-1}$			0.071			0.070
			(0.070)			(0.070)
DS $\text{Ten.}_{t-1} \times \text{HD}_{t-1}$				-0.009		-0.006
				(0.029)		(0.029)
DS $\text{Ten.}_{t-1} \times \text{PD}_{t-1}$					0.043	0.040
					(0.098)	(0.100)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2942	2955	2942	2942	2942	2942
Adj. R^2	0.616	0.626	0.616	0.616	0.616	0.616

Notes: (1) Lagged strategy only (Eq. 2); (2) Current and lagged strategy (Eq. 3); (3) Eq. 2 + BS Tenure_{t-1} × Hybrid B_{t-1} (Eq. 4); (4) Eq. 2 + DS Tenure_{t-1} × Hybrid D_{t-1} (Eq. 5); (5) Eq. 2 + DS Tenure_{t-1} × Pure D_{t-1} (Eq. 6); (6) All three tenure interactions simultaneously. HC1 robust SE. Firm and year FE throughout. Omitted baseline: Pure B. Controls (current and lagged): ln(Capital), ln(Labor), PPE, Total Employment. Lagged variables: $t-1$ within firm. BS Tenure: `ndstenure` (non-data-specialist tenure); DS Tenure: `dstenure` (data-specialist tenure). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

All Hybrid B coefficients are statistically insignificant across all six columns. In col. 1 (lagged-only), $\text{HB}_{t-1} = -0.055$ (s.e. 0.052). In col. 2 (current+lagged), both HB (+0.052, s.e. 0.057) and HB_{t-1} (-0.070, s.e. 0.053) are insignificant. The BS_Tenure × HybridB interaction (col. 3: +0.071, s.e. 0.070) is insignificant; the BS_Tenure main effect (+0.242**, s.e. 0.119) is the only significant term, indicating longer-tenured business specialists are associated with higher ROA but independently of Hybrid B status.

Implication: There is no evidence of delayed or cumulative Hybrid B effects. Firms with long-tenured business specialists do not earn a Hybrid B premium from data integration. The ROA benefit of tenure accrues to business specialists irrespective of whether their firm adopts Hybrid B.

3.5 Task 3E: Instrumental Variables / Shift-Share IV

Construction and design: A shift-share (Bartik) instrument is constructed to address reverse causality and omitted-variable bias in the Hybrid B – ROA relationship. The design exploits the interaction between a firm's pre-period occupation-posting exposure (the "share" component, fixed at baseline year 2011) and national trends in data-skill adoption across onet segments (the "shift" component). Because the 2011 posting mix pre-dates the outcome window, the share is plausibly predetermined; the national shift is made exogenous to any individual firm by computing it on a leave-one-out basis that excludes firm i's own postings:

$$Z_{i,t} = \sum_{s} \hat{\sigma}_{i,s,2011} \times \text{LOO-Share}_{s,t}^{-i} \tag{7}$$

where $\hat{\sigma}_{i,s,2011}$ is firm i's share of business postings in onet segment s at baseline (2011), and $\text{LOO-Share}_{s,t}^{-i}$ is the leave-one-out national HybridB posting share for onet s in year t. The instrument is constructed from a single pass through `SP500_jobs_20102023.csv` (9.0 min runtime for 104 chunks × 500K rows). The resulting instrument covers 3,916 of 4,217 firm-years (92.9%); mean $Z = 0.565$, s.d. $= 0.110$.

The instrument enters a two-stage least squares (2SLS) procedure, estimated separately for two endogenous variables: the binary Hybrid B dummy and HybridB_Intensity (the posting-flow continuous measure from Task 1A). Firm and year fixed effects are absorbed via sequential within-group demeaning prior to estimation; the reported first-stage F-statistic is the squared t-ratio on $Z_{i,t}$ in the demeaned first-stage regression. The first stage (Eq. 8) is estimated under Firm+Year FE:

$$\text{HybridB}_{i,t} = \pi Z_{i,t} + \gamma \mathbf{X}_{i,t} + \mu_i + \lambda_t + u_{i,t} \tag{8}$$

The second stage (Eq. 9) recovers the 2SLS estimate:

$$\ln(\text{ROA})_{i,t} = \beta \widehat{\text{HybridB}}_{i,t} + \gamma \mathbf{X}_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \tag{9}$$

Table 18: First Stage: Shift-Share IV (Task 3E)

	(1) HybridB	(2) HybridB Intensity
Shift-Share IV ($Z_{i,t}$)	0.012	0.038
	(0.087)	(0.055)
ln_capital	-0.089***	0.007
	(0.020)	(0.011)
ln_labor	0.111***	-0.017
	(0.031)	(0.014)
ppe	0.003***	0.000
	(0.001)	(0.000)
totemp	-0.000	-0.000*
	(0.000)	(0.000)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	3184	3184
First-stage F (instrument)	0.02	0.47

Notes: HC1 robust SE. Firm and year FE absorbed via sequential within-group demeaning. Dependent variable: Col. 1 = Hybrid B dummy; Col. 2 = HybridB Intensity. Instrument $Z_{i,t} = \sum_s \hat{\sigma}_{i,s} \cdot$ LOO-Share$_{s,t}$, where $\hat{\sigma}_{i,s}$ is firm iś baseline (2011) share of business postings in onet s, and LOO-Share$_{s,t}$ is the leave-one-out national HybridB posting share for onet s in year t. $F < 10$ flags a weak instrument. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The first-stage F-statistics are 0.02 (Hybrid B binary) and 0.47 (HybridB_Intensity), far below the conventional weak-instrument threshold of 10. The instrument has essentially no predictive power for Hybrid B adoption. This occurs because national data-skill adoption trends are nearly uniform across onet segments—within-onet variation in HybridB share is low and similar across all firms, so firm-level baseline shares do not differentiate the instrument.

Table 19: Second Stage: 2SLS Estimates (Task 3E)

	(1) Hybrid B	(2) HybridB Intensity
Hybrid B (instrumented)	14.228	
	(100.278)	
HybridB Intensity (instr.)		4.636
		(8.142)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	3184	3184
First-stage F	0.02	0.47

Notes: 2SLS via shift-share instrument. Robust SE. Firm and year FE absorbed via sequential demeaning. Col. 1: endogenous = Hybrid B dummy. Col. 2: endogenous = HybridB Intensity (posting-flow Eq. 1). $F < 10$ flags a weak instrument. Controls: ln(Capital), ln(Labor), PPE, Total Employment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The 2SLS second-stage estimates are implausibly large: HybridB = +14.2 (s.e. 100.3) and HybridB_Intensity = +4.6 (s.e. 8.1), reflecting classic weak-instrument bias and offering no interpretable information.

Implication: The shift-share instrument is underpowered for this application. The IV design cannot provide causal inference here; the weakness of the instrument is itself an informative finding, suggesting that national trends in data-skill demand are too homogeneous across business occupations to generate plausibly exogenous firm-level variation.

4 Conclusion

4.1 Overall Assessment

The 13 robustness and causal-validation tasks deliver a coherent conclusion.

1. **Cross-sectional association (between-firm):** The Hybrid B premium of $+0.227^{***}$ log-points (industry+year FE) is a robust cross-sectional correlation. It survives industry×year FE ($+0.219^{***}$), 4-digit NAICS FE ($+0.131^{***}$), and firm-clustered standard errors ($+0.219^{**}$), ruling out broad industry trends and time aggregation as explanations. The attenuation from +0.227 to +0.131 at the 4-digit NAICS level indicates partial within-narrow-industry sorting, but the premium remains large and significant.

2. **Within-firm causal effect:** Once firm fixed effects are introduced, the coefficient falls to +0.030 (s.e. 0.049, $p > 0.5$) and remains indistinguishable from zero across every tested variant: alternative matching methods, staggered DiD estimators, lagged outcomes, alternative DV transformations, persistence restrictions, and single-transition samples. The Sun–Abraham (−0.090) and Callaway–Sant'Anna (−0.050) staggered estimators produce *more* negative estimates than TWFE (−0.023), ruling out heterogeneous-treatment-effect bias as an explanation for the null.

3. **Interpretation:** The evidence is consistent with *positive selection into Hybrid B*: firms that adopt the strategy are systematically better-performing (on unobserved dimensions) before adoption. The PSM event study (Task 2A) is the cleanest test: after matching on pre-adoption ROA trajectory and all observable covariates, the post-adoption effect is −0.002. The FE ladder (Task 3B) confirms the mechanism is entirely between-firm. Persistent adopters (Task 2C, $N = 39$) show the most negative post-adoption trajectory (avg post = −0.174), ruling out transitory adoption noise as the source of the null.

4.2 Conclusion Table

Table 20 summarises the evidence from all 13 tasks. "Supports main result?" refers to whether the task finds a positive, significant Hybrid B effect. "Supports causal interpretation?" refers to whether the identification strategy supports attributing any estimated effect to the Hybrid B strategy itself.

Table 20: Conclusion Table: Evidence Summary Across All Tasks

Task	**Analysis**	**Main result?**	**Causal?**	**Notes**
Task 1A	Continuous intensity measures	No	N/A	Intensity null in all specs. Binary premium not replicated.
Task 1B	Percentile-based thresholds	No	N/A	Top 20–40% cutoffs: Firm FE coefs −0.018 to +0.033, all n.s.
Task 2A	Pre-trend-adjusted matching	No	No	60/61 matched; pre-trend $p = 0.39$; avg. post = −0.002 (n.s.).
Task 2B	Alternative matching & weighting	No	No	PSM: −0.002, −0.022 (n.s.). IPW/EB positive but composition-driven.
Task 2C	Persistent Hybrid B adoption	No	No	Persistent adopters ($N = 39$): avg. post = −0.174. Worse than non-persistent.
Task 2D	Clean Pure B-to-Hybrid B transition	No	No	Pure B→HB: avg. post = −0.013 (n.s.). HD overlap not confounding.
Task 2E	Staggered DiD estimators	No	No	TWFE = −0.023; SA = −0.090; CS = −0.050. Heterogeneity bias not an issue.
Task 2F	Lagged-outcome models	No	No	t+1 = −0.041, t+2 = +0.045, both n.s. No delayed effect.
Task 3A	Strategy persistence, single-trans.	No	N/A	HB coefs −0.073 to +0.030 across 4 specs, all n.s.
Task 3B	Fine-grained fixed effects	Partial	N/A	Cross-sec.: +0.219–+0.227***; Firm FE: +0.030 (n.s.). Between-firm only.
Task 3C	Alternative performance measures	No	N/A	5 DV transforms; all HB coefs n.s.
Task 3D	Lagged dynamics & tenure	No	N/A	No lagged HB effect. BS Tenure independent of HB.
Task 3E	Shift-share IV	No	No	First-stage $F < 1$; 2SLS uninformative. Weak instrument.

Notes: "Main result?" = does this task find a positive, statistically significant Hybrid B effect? "Causal?" = does the identification strategy support a causal interpretation (No / Partial / Yes)? N/A = task design does not speak to causality. All within-firm results use Firm+Year FE; cross-sectional results use Industry+Year FE. HC1 robust standard errors unless stated otherwise. Preferred threshold throughout: $\tau = 0.20$. Omitted baseline: Pure B.

4.3 Final Remarks

Three tiers of evidence emerge from this analysis.

1. **Tier 1 — Robust cross-sectional association:** Task 3B confirms the $+0.227^{***}$ premium survives industry×year FE ($+0.219^{***}$), 4-digit NAICS FE ($+0.131^{***}$), and firm-clustered standard errors ($+0.219^{**}$). The between-firm correlation is real and not an artefact of broad industry trends or time aggregation.

2. **Tier 2 — Selection, not causation:** Six causal designs (Tasks 2A–2F) consistently find average post-treatment effects near zero or slightly negative, with parallel pre-trends in every case. The divergence between PSM methods (-0.002, -0.022) and IPW/EB ($+0.185$, $+0.254$) is direct evidence of selection confounding: only when treated and control firms are matched like-for-like does the premium vanish. Tasks 3A and 3D confirm the null under persistence restrictions and lagged specifications; Task 3D additionally shows the ROA benefit of business-specialist tenure accrues independently of Hybrid B status, pinpointing the selection to firm quality rather than hiring composition.

3. **Tier 3 — Underpowered or uninformative:** Tasks 1A and 1B show the premium does not generalise to posting-flow intensity measures, limiting the construct validity of the Stata binary classification. Task 3E's shift-share instrument is too weak (first-stage $F < 1$) because national data-skill adoption trends are too uniform across business occupations to generate plausibly exogenous firm-level variation.

In summary: the Hybrid B strategy is associated with better-performing firms, but the available evidence does not support the interpretation that adopting Hybrid B *causes* firms to perform better. The causal question remains open and would require either a natural experiment generating plausible exogenous variation in Hybrid B adoption, or panel data extending beyond 2019 to increase the post-adoption observation window.